Course Code: CEF 440

Course Name: Internet Programming and Mobile Programming

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GROUP 11

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Task 2: Requirement Gathering

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Introduction

Modern vehicles use dashboard warning lights and engine sounds to indicate faults, but many car owners do not understand these signals. This often leads to delayed maintenance, higher repair costs, and reduced vehicle lifespan. Existing vehicle diagnostic tools are usually expensive, complex, or require special hardware, making them difficult for ordinary users to access.

With the growth of Artificial Intelligence (AI) and mobile technology, smartphones can now be used to help diagnose vehicle problems. This project focuses on designing and implementing an AI-based mobile application that can scan dashboard warning lights using the phone camera and analyze engine sounds using audio recognition techniques.

The application will provide users with fault explanations, urgency levels, repair suggestions, maintenance tips, and video tutorials from car experts. It will also support both offline and online features to improve accessibility and convenience. The main goal of the system is to help car owners detect vehicle problems early, reduce maintenance costs, and improve vehicle maintenance through smart mobile technology.

A. Stakeholder Identification

Stakeholders are individuals, groups, or organizations that are directly or indirectly affected by the development and use of the mobile car fault diagnosis system.

1. Car Owners / Drivers (Primary Stakeholders)

These are the main users of the application. They use the system to:

- Scan dashboard warning lights.
- Record engine sounds.
- Receive fault explanations and repair recommendations.
- Access maintenance tutorials and guidance.

Their feedback is very important because the application is designed to solve their everyday vehicle maintenance challenges.

2. Mechanics and Automotive Technicians

Professional mechanics provide expert knowledge about:

- Common vehicle faults.
- Dashboard warning indicators.

- Engine sound patterns.
- Recommended repair procedures.

They may also validate the accuracy of the diagnoses produced by the system and contribute training data for the AI models.

3. System Developers

These include software engineers, mobile app developers, AI engineers, and database designers responsible for:

- Designing the application architecture.
- Developing AI models for image and sound recognition.
- Implementing the mobile application features.
- Testing and maintaining the system.

4. Vehicle Maintenance Experts / Content Creators

These stakeholders provide educational repair and maintenance content such as:

- Video tutorials.
- Step-by-step repair demonstrations.
- Maintenance advice and safety tips.

Their content helps users understand how to solve or manage detected faults.

5. Automobile Companies and Diagnostic Data Providers

Manufacturers and automotive data providers may contribute:

- Information about dashboard warning symbols.
- Fault code databases.
- Vehicle maintenance standards.
- Technical documentation for different car models.

6. Internet Service Providers / Cloud Service Providers

Online features such as updated fault databases, cloud storage, and video streaming depend on internet and cloud services. These stakeholders help ensure:

- Reliable data access.
- Fast content delivery.
- Secure online functionality.

7. Project Supervisor / Academic Institution

The project supervisor and institution guide the project by:

- Approving project objectives.
- Monitoring progress.
- Evaluating system quality and documentation.
- Ensuring academic standards are met.

B. Requirement Gathering Techniques

Requirement gathering is the process of collecting information about the system's features, user needs, and expected functionalities. For this project, the following techniques can be used:

1. Interviews

Interviews involve direct discussions with stakeholders such as car owners and mechanics to understand their challenges and expectations.

1. Purpose in This Project

· To understand common vehicle problems faced by users.

· To identify the most confusing dashboard warning lights.

· To gather expert knowledge about engine sound diagnosis.

· To know what features users expect in the application.

2. Interview participant

· 4 professional drivers

· 2 truck drivers

· 2 students who drive frequently

· 4 Car Repairer (Mechanic)

· 2 lecturer that drives

3. Interview Format

· Semi-structured interviews (a mix of guided and open discussion)

· Conducted in person and voice calls via WhatsApp

4. Sample interview Question and Responses

Question:

How important is it for you to receive guidance (like video tutorials or repair suggestions) directly through a mobile app when diagnosing car problems?

Responses:

Driver1: “Very important, especially for inexperience drivers and for me it might ease repair”

Driver 2: “Not interested in the guideline and because the company will send a mechanic that u can’t trust”

Driver 3: “Yes normally as the world is changing, we need to follow the trend, don’t know how to use android very well but if you make a good app, it my make me to learn it”

Question:

What other information can you tell us concerning dash board warning lights and sounds detection

Answer

Driver 1 -When you see the battery light on during driving it means the generator or the battery has an issue. -when the fuel gauge is at F it means that the tank is full, when it’s at E it means the fuel tank is empty.

Driver 2 “When the modification light is on during motion it means the modio has a fault”

Here are some pictures in action

5. Its Advantages to us

- Provides detailed and accurate information.
- Allows clarification of responses.
- Helps gather expert opinions from mechanics.

2. Questionnaires / Surveys

Questionnaires are written sets of questions distributed to many users to collect information quickly.

1. Purpose in This Project

- To gather opinions from a large number of car owners.
- To identify the most common maintenance challenges.
- To determine how often users experience warning light confusion.
- To evaluate user interest in AI-based diagnostics.

2. Target Audience

· Regular car owners with minimum and maximum technical knowledge(40 Responses)

· Students with vehicles either cars or bike

· Delivery drivers

· Truck driver

· Another survey for Car Repairer (Mechanic, 12Responses)

3. Survey Design

The survey was distributed online (via google forms). It includes a mix of:

· **Multiple- choice question** to gather measurable data

· **Open-Ended questions** to allow additional suggestion and feedback

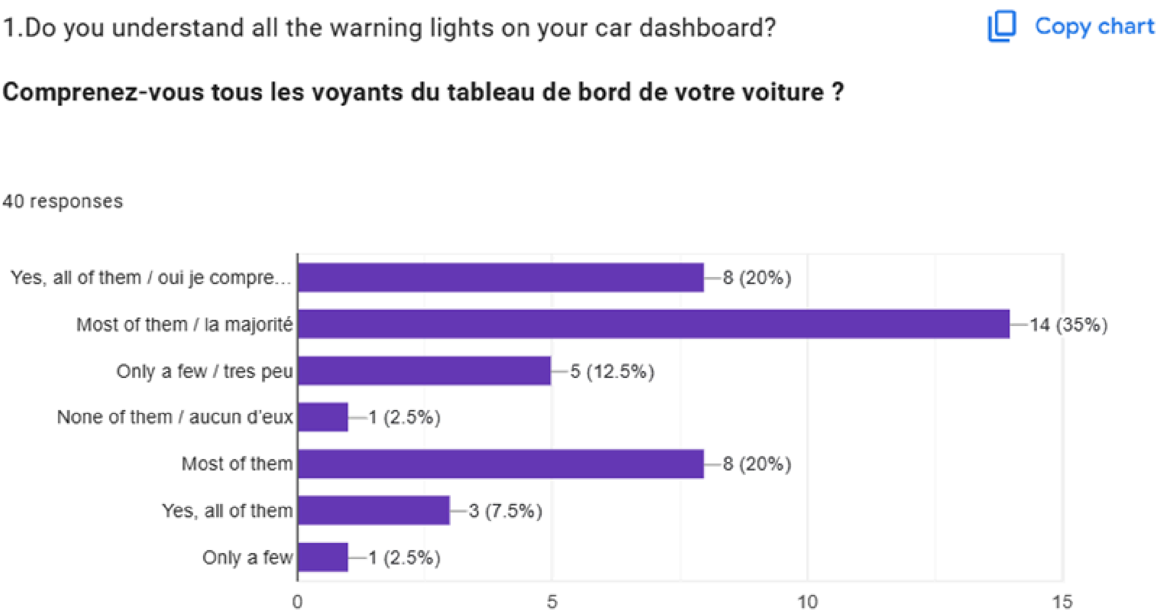
· **Multi language** because since Cameroon is a bilingual country it will encourage both parties to be able to answer without issues

4. . Sample Questions and Key Findings from the Google form for Car Owner

1.Do you understand all the warning lights on your car dashboard?

Comprenez-vous tous les voyants du tableau de bord de votre voiture ?

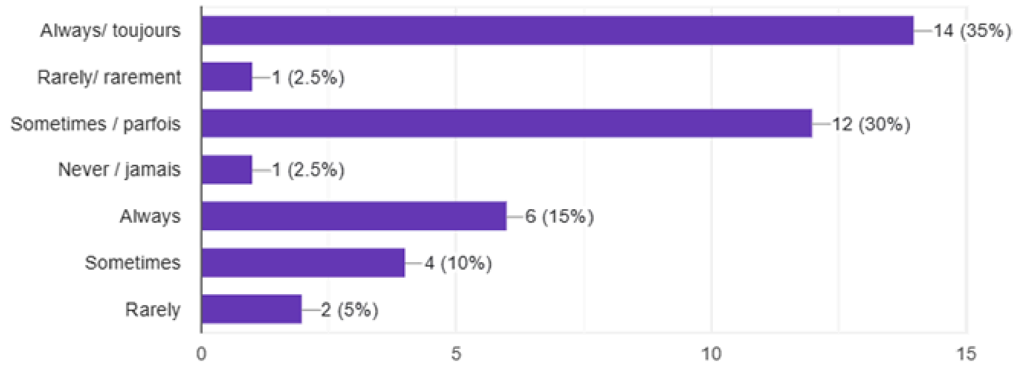
40 responses



4. Do you visit a mechanic immediately when your car has a problem? / **Vas-tu chez un mechanicien directement après avoir vu une faille ?**

Copy chart

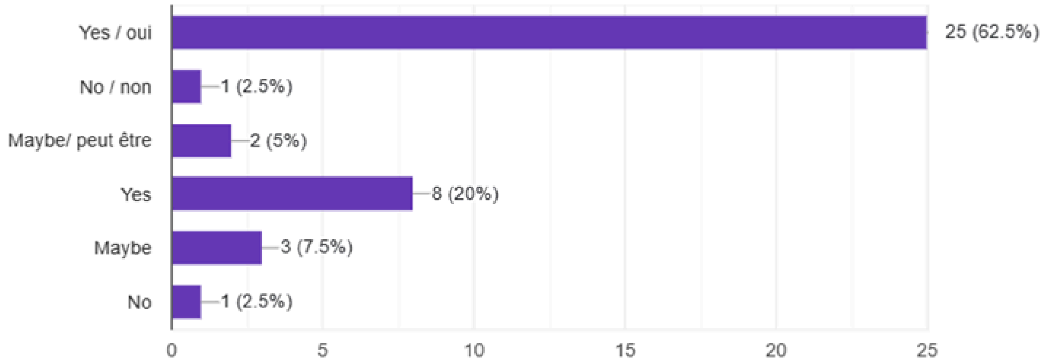
40 responses



7. Would you like an app that explains car problems using your phone?/**Souhaiteriez-vous une application mobile qui explique les problèmes de voiture à l'aide de votre téléphone ?**

Copy chart

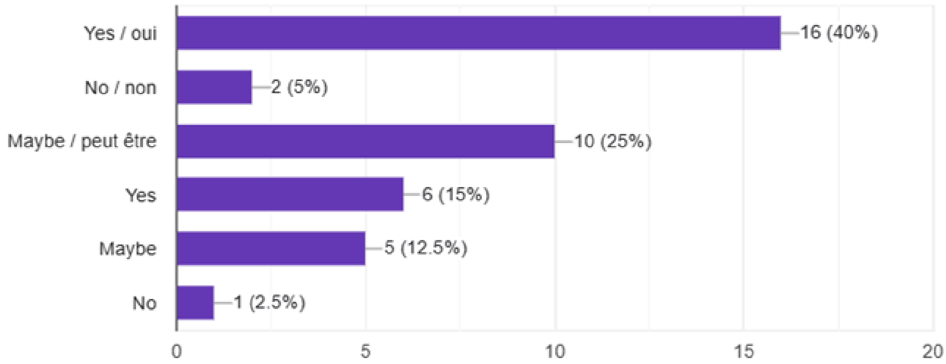
40 responses



8. Would you trust an app to diagnose your car issues? /**Feriez-vous confiance à une application pour diagnostiquer les problèmes de votre voiture ?**

Copy chart

40 responses



Why?(Optional) /si oui pourquoi ?

28 responses

He can help detect potential issues with my car in advance.

I'm into tech so I'll trust any form of new initiative that can help diagnose car problems

Even after diagnosis it will end up repaired by a professional

Not optional

To avoid paying more than what can solve the problem to the technician

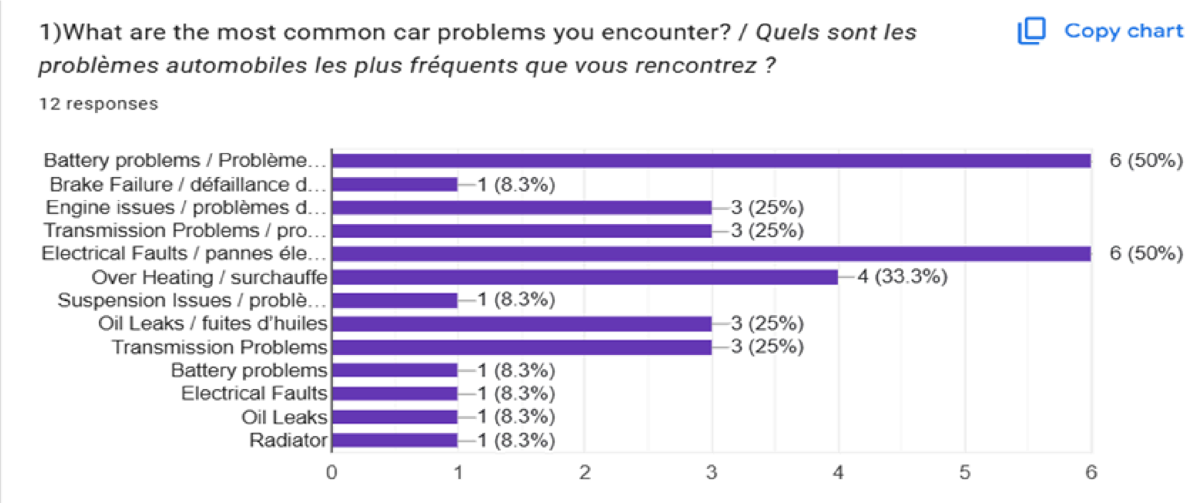
Kilo hadison

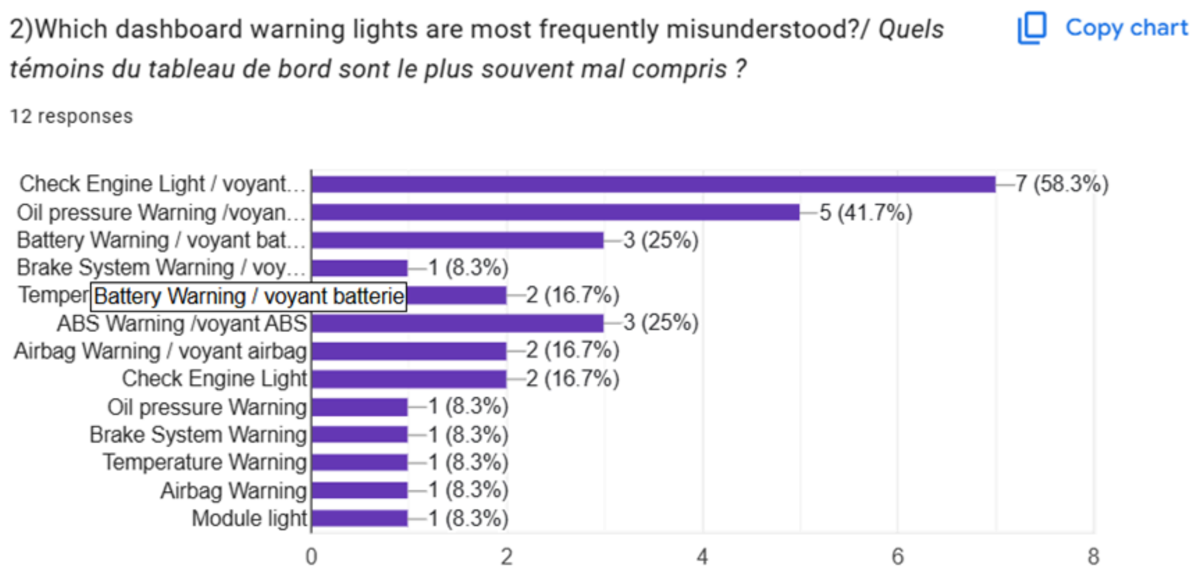
Some mechanics do it wrongly

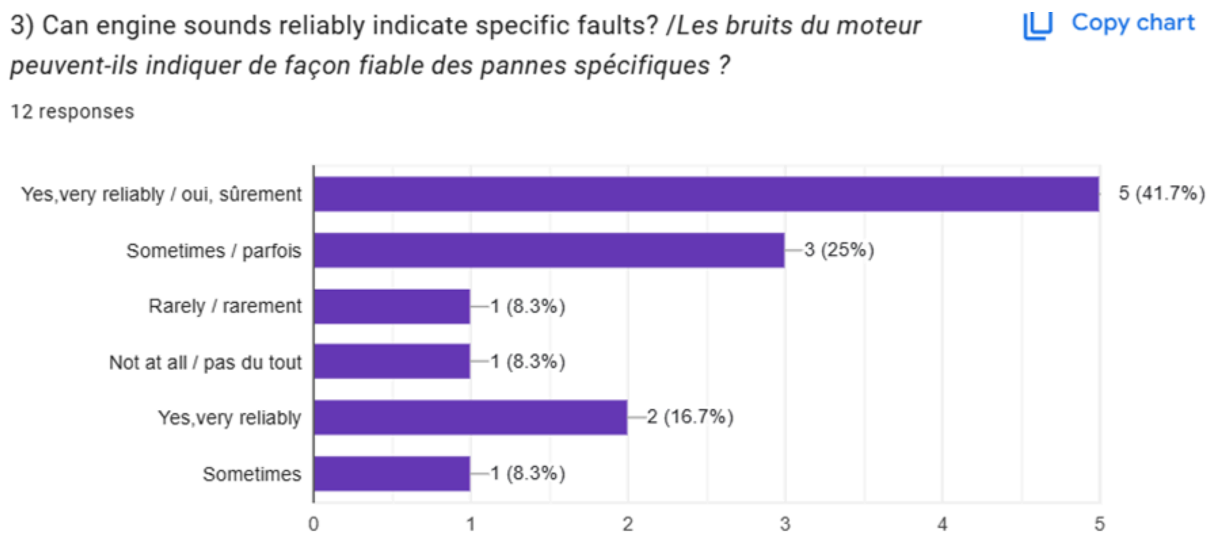
Makes it easier

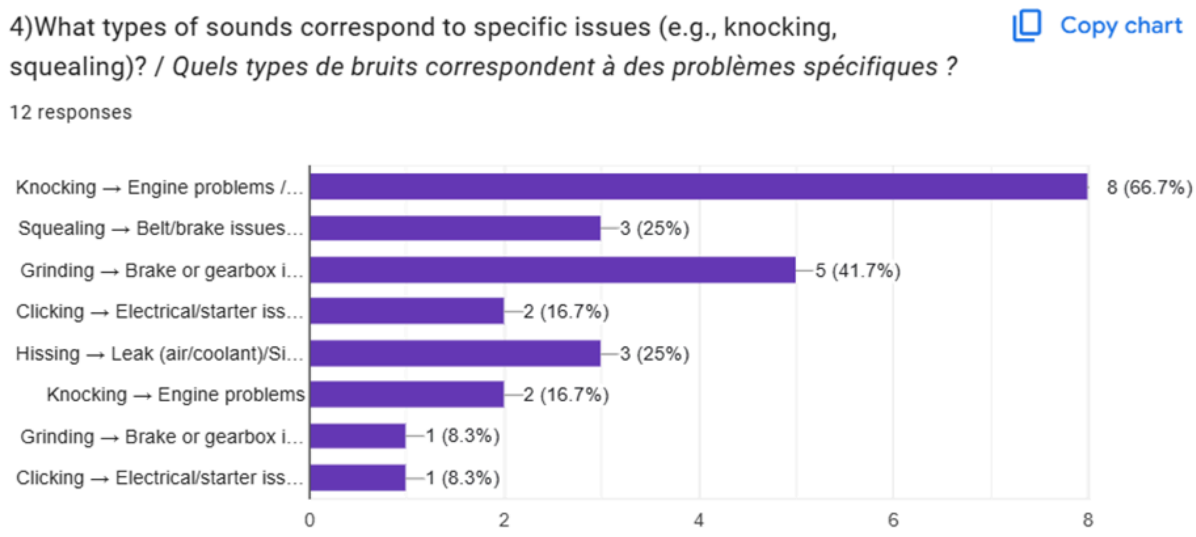
Some want to scam u

5. Sample Questions and Key Findings from the Google form for Car Repairer









6. Its Advantages to Us

- Reaches many participants quickly.
- Saves time and cost.
- Produces data that can easily be analyzed statistically.

C. Brainstorming

Brainstorming is a creative group technique used to generate a broad range of ideas, solutions, and insights around a central theme or problem. For the development of our mobile application aimed at diagnosing car faults using AI, we conducted structured brainstorming sessions to gather functional and non-functional requirements from stakeholders.

1. Objective of the Brainstorming Session

The main goals of the brainstorming session were:

· To define the key features and functionalities of the application.

· To decide on the type of mobile application (native, hybrid, or cross-platform).

· To choose an appropriate development framework and supporting technologies

2. Participants

The brainstorming session was attended by all members of the student project team. Each member contributed ideas based on their user expectations and current trends in mobile app development.

3. Method

We used a free-form group discussion approach where every team member shared their ideas openly. Notes were taken on a piece of paper, and ideas were grouped into themes:

· App Features

· App Type & Compatibility

· Framework & Tools

4. Key Points Discussed

A. App Features

We identified several cores features the app should have to assist users in diagnosing car faults:

1. Dashboard Light Recognition

· Use the phone's camera to scan dashboard warning lights. o Provide explanations for each symbol with urgency levels.

2. Engine Sound Diagnosis

· Use the phone microphone to record engine noise.

- Analyze sound patterns using AI to identify issues.
- Match sounds with a trained dataset to provide likely fault causes.

3. User Guidance and Recommendations

- Show possible causes and repair suggestions.
- Include links to video tutorials from experts (e.g., YouTube integration).
- Prioritize issues based on urgency (e.g., Immediate, Moderate, Minor).

4. Offline and Online Functionality

- Basic diagnostic features should work offline using preloaded data.
- Online features will allow access to updated fault databases and tutorial content.

5. History and Records

- Save past scans and sound diagnoses for user reference.

B. Type of Application

We agreed to build a cross-platform mobile application to ensure compatibility with both Android and iOS devices. This will help reach a wider user base and reduce the workload of maintaining two separate codebases.

D. **Reverse Engineering**

Reverse engineering was applied as a technique by examining currently available car diagnostic applications and their features, interfaces, and limitations. By analysing how these existing tools operate , what they can and cannot do, how they present information to users, and where they fall short for non-technical car owners , it was possible to derive implicit requirements that stakeholders might not have articulated directly.

1. Reverse Engineering of Existing Applications

Two similar applications were studied to understand:

- Their core functionalities
- How they interact with users
- Their strengths and weaknesses

2. Problem Analysis

The difficulties faced by car owners were analyzed, including:

- Inability to interpret dashboard warning lights
- Difficulty identifying engine faults based on sound
- Dependence on mechanics for minor issues

3. Feature Extraction and Gap Identification

Features from existing systems were extracted and compared with user needs to identify missing functionalities.

A. FIXD OBD2 Scanner App

Overview:
This application provides vehicle diagnostics by connecting to the car through an external OBD2 device.

Key Features:

- Reads vehicle fault codes
- Provides simple explanations of issues
- Suggests possible fixes
- Indicates urgency levels

Limitations:

- Requires external hardware (OBD2 scanner)
- Does not support image-based detection
- No engine sound analysis capability

B. Car Scanner ELM OBD2

Overview:
This application focuses on real-time vehicle monitoring and diagnostics using sensor data.

Key Features:

- Displays real-time engine data
- Provides diagnostic trouble codes

- Visualizes data using charts

Limitations:

- Complex interface for non-technical users
- Requires external hardware
- No AI-based automation
- No support for image or audio-based diagnosis
-

4. Identified Gaps in Existing Systems

From the analysis, the following gaps were identified:

- Lack of **computer vision capabilities** for dashboard light recognition
- Absence of **audio-based fault detection**
- Dependence on additional hardware devices
- Limited accessibility for non-technical users
- Lack of integrated multimedia guidance (e.g., repair videos)

2. DATA GATHERING

Introduction

To ensure a comprehensive understanding of the problem domain, requirement gathering for this project was conducted using **two complementary surveys**:

- A **Car Owner Survey** to capture user experiences, challenges, and expectations
- A **Mechanic Insight Survey** to gather expert knowledge on vehicle faults, diagnosis methods, and practical limitations

This dual approach ensures that the proposed system is both **user-centered** and **technically reliable**.

2.1 Requirement Gathering Approach

The following methods were used:

1. User Survey (Car Owners)
Focused on:

- Understanding user difficulties
- Identifying behavior when faults occur
- Determining desired features and usability needs

2. Expert Survey (Mechanics)
Focused on:

- Common vehicle faults and warning signs
- Reliable diagnostic methods
- Real-world challenges in car maintenance
- Practical limitations of automated diagnosis

Car Owner Survey Summary
The questionnaire consisted of **12 structured questions**, divided into the following categories:

1. Knowledge of Car Faults
Questions 1, 3

- Understanding of dashboard warning lights
- Reasons for ignoring warnings

2. User Behavior and Habits
Questions 2, 4

- Actions taken when warning lights appear
- Frequency of consulting mechanics

3. Experience with Engine Sounds
Questions 5, 6

- Frequency of unusual sounds
- Types of sounds experienced (knocking, squealing, etc.)

4. Acceptance of Mobile Solutions
Questions 7, 8

- Willingness to use a mobile app
- Level of trust in automated diagnosis

5. Desired Features
Question 9, 10

- Expected functionalities (e.g., video tutorials, sound detection)
- Preference for video-based guidance

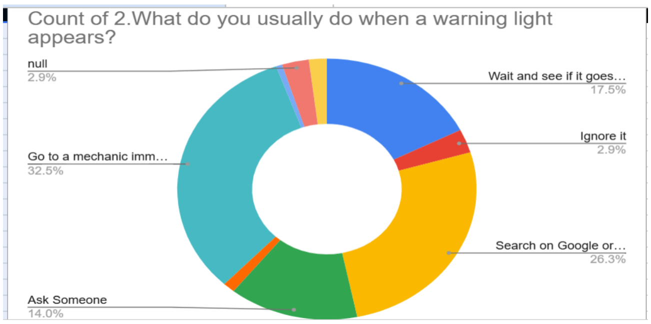
6. Technical Constraints
Questions 11, 12

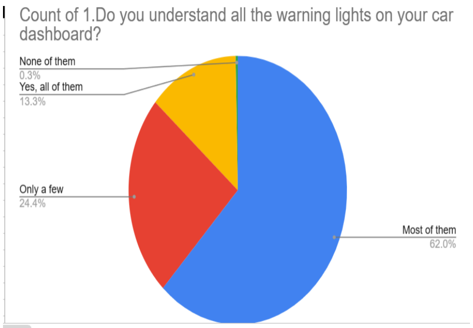
- Internet availability
- Expected ease of use

Key Findings from the Survey (Interpretation)

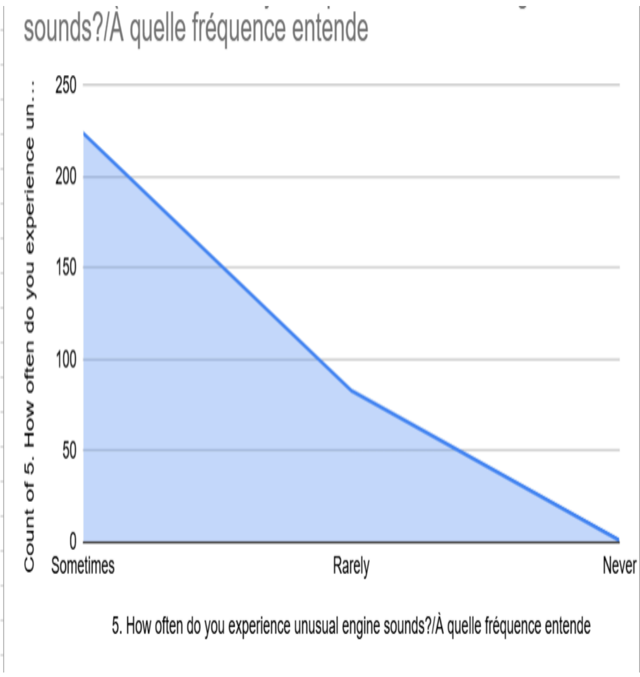
The analysis of responses revealed the following trends:

- 62% of users **do not fully understand dashboard warning lights**, with many indicating “only a few” or “none.”
- Approximately 60% users rely on **Google, friends, or mechanics**, rather than understanding the issue themselves.



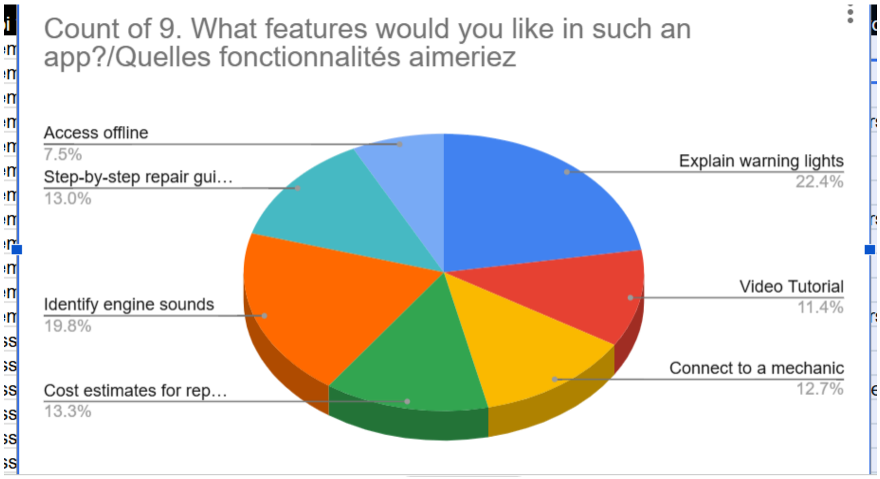


- Some users admitted to **ignoring warning lights**, mainly due to lack of knowledge or assuming the issue is not serious.
- A considerable number of respondents reported experiencing **unusual engine sounds**, such as knocking and squealing.



- Most participants expressed **interest in a mobile application** that can diagnose car problems.
- However, trust in such applications is **moderate**, indicating the need for reliability and clear explanations.
- The most requested features include:

- Explanation of warning lights
- Engine sound identification
- Video tutorials
- Step-by-step repair guides



- Many users do not have constant internet access, highlighting the importance of **offline functionality**.

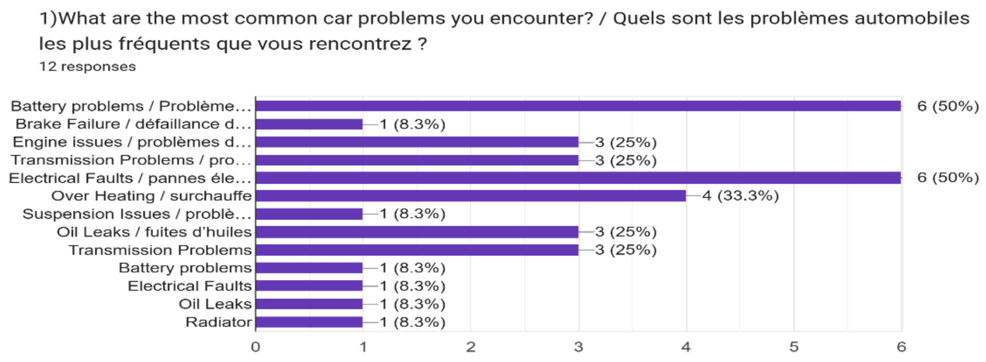
Mechanic Survey Design and Structure

Survey Analysis

A. Common Car Problems

Mechanics identified the most frequent issues as:

- Engine problems
- Battery faults
- Brake system failures
- Electrical issues
- Overheating and oil leaks



Implication:

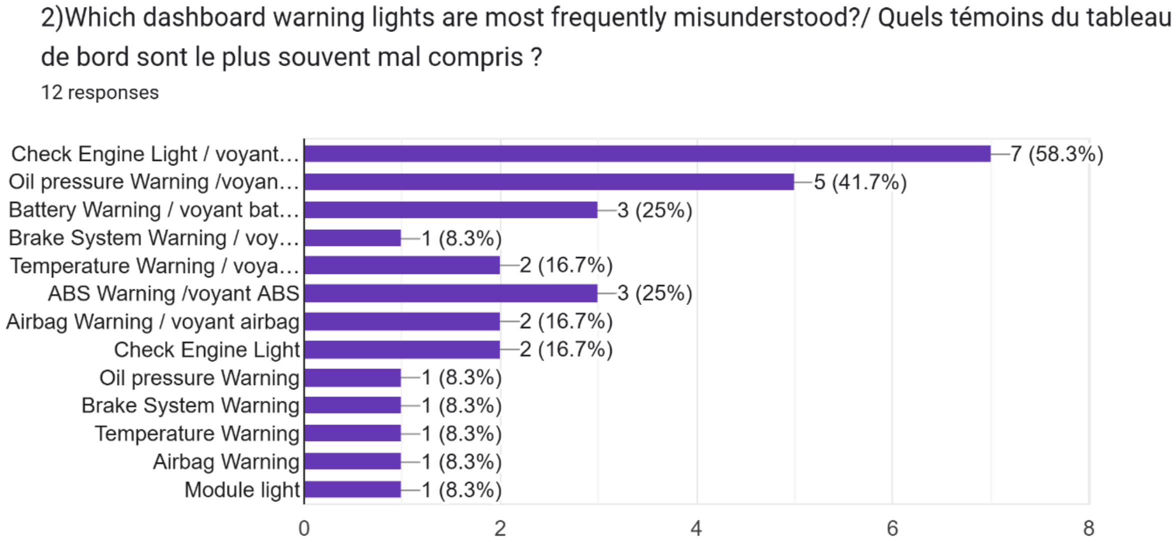
The system must prioritize detection of these common faults.

B. Misunderstood Warning Lights

The most confusing indicators include:

- Check engine light

- Oil pressure warning
- Battery warning
- Brake system warning



Implication:
These warning lights should be **clearly explained in the app**, with causes and urgency.

C. Reliability of Engine Sounds

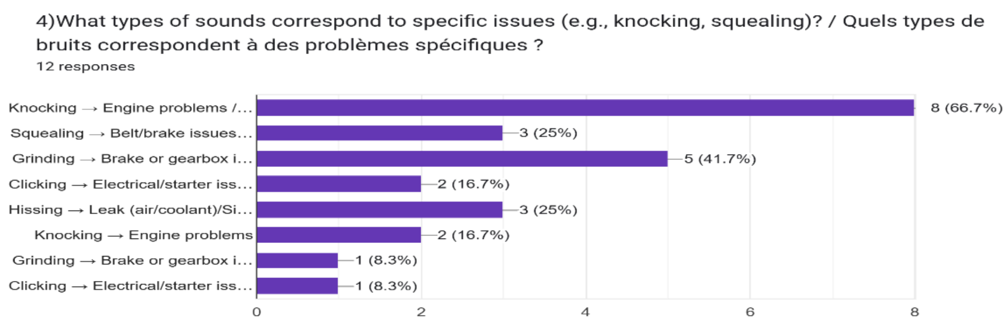
- Many mechanics confirmed that **engine sounds can indicate faults**, though not always perfectly

Implication:
Audio diagnosis should be used as a **supporting feature**, not the only decision-maker.

D. Sound–Fault Mapping

Mechanics provided clear mappings:

- Knocking → engine problems
- Squealing → belt or brake issues
- Grinding → gearbox or brake issues
- Clicking → electrical/starter issues
- Hissing → leaks

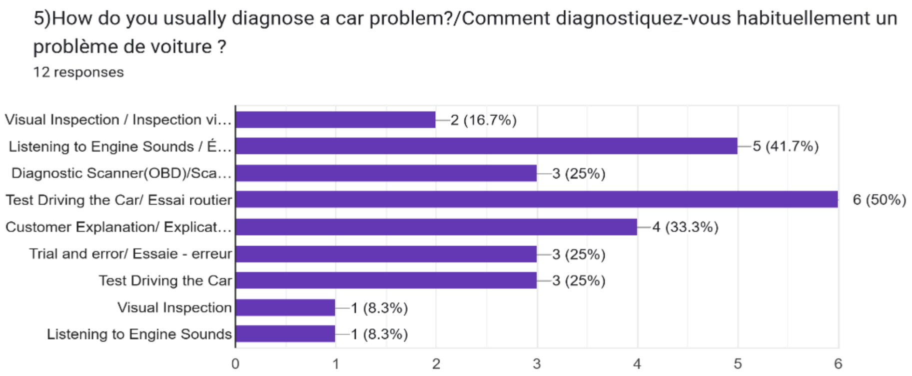


Implication:
These mappings can be used to **train the audio classification model**.

E. Current Diagnosis Methods

Mechanics rely on:

- Visual inspection
- Listening to sounds
- Test driving



Implication:
Your app should simulate:

- **Vision (camera)**
- **Hearing (microphone)**

F. Identified Limitations

Mechanics highlighted key risks:

- Misdiagnosis
- Lack of physical inspection
- Device limitations (microphone quality)
- User misunderstanding instructions

Implication:
The app must:

- Include disclaimers
- Provide confidence levels
- Keep instructions simple

G. Repair Guidance

Mechanics confirmed that users can handle simple tasks like:

- Changing battery
- Replacing oil
- Changing tyres
- Refilling coolant

Implication:
The app should include **basic repair tutorials only**, not complex repairs.

Accuracy Expectations

Most mechanics expect **high or very high accuracy**

Implication:
The system must prioritize **reliability and correctnessCommon User Mistakes**

- Ignoring warning lights
- Delaying repairs
- Attempting complex fixes
- Poor maintenance

Implication:
The app should:
- Educate users
- Warn against risky actions

Combined Findings
By combining both surveys:

Car Owners Need	Mechanic Insight	System Decision
Don’t understand warning lights	Certain lights are often misunderstood	Add detailed explanations + urgency
Hear sounds but don’t understand	Sounds can indicate faults	Add audio recognition
Want simple help	Diagnosis is complex	Simplify results
Want tutorials	Only basic repairs are safe	Provide basic video guides
Limited internet	Diagnosis tools are technical	Add offline AI features

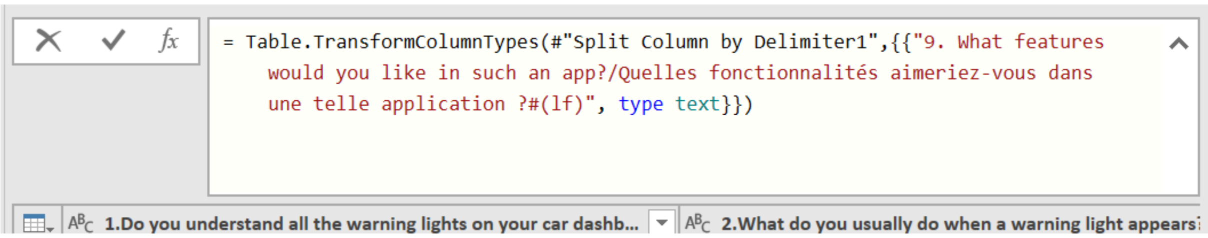
DATA CLEANING

Following the collection of responses from both surveys, a data cleaning process was carried out to ensure that the dataset was accurate, consistent, and suitable for analysis. Raw survey data collected through Google Forms often contains irregularities arising from bilingual response options, multiple-selection questions, and variations in how respondents interpret and answer questions. The following cleaning steps were applied to both datasets before the data was analysed and used to derive system requirements.

- **Consolidation of Bilingual Responses**

  The most significant source of inconsistency in both datasets was the bilingual nature of the survey questions and answer options, which were presented in both English and French. Because Google Forms treats the English version and the French version of the same answer as two distinct entries, the raw data contained duplicate categories that represented identical responses. For example, the answer "Most of them" and "Most of them / la majorité" were recorded as separate values despite conveying the same meaning. Similarly, "Yes" and "Yes / oui", "Rarely" and "Rarely / rarement", and all equivalent bilingual pairs across both surveys were merged into single unified categories during cleaning. This consolidation was applied systematically to every question in both surveys before any frequency counts or percentages were computed, ensuring that the resulting statistics accurately reflect the true distribution of responses.

  To go about this we used the split formula in the data top ribbon of excel



After this we got the following result

5. How often do you experience unusual e	6. Can you describe any sound your car ha	7. Would you like an app that explains car
Sometimes / parfois	Clicking/Cliquetis	Yes / oui
Never / jamais	None	Yes / oui
Sometimes / parfois	Knocking/ cognement, Squealing /grincement	Yes / oui
Sometimes / parfois	Squealing /grincement	Yes / oui
Sometimes / parfois	Knocking/ cognement, Hissing/ sifflement	Yes / oui
Sometimes / parfois	I haven't noticed anything / je n'ai jamais preté	Maybe/ peut être
Sometimes / parfois	Squealing /grincement	Yes / oui
Sometimes / parfois	Squealing /grincement	Yes / oui
Rarely / rarement	I haven't noticed anything / je n'ai jamais preté	Yes / oui
Sometimes / parfois	Grinding / frottement métallique	Yes / oui
Sometimes / parfois	Clicking/Cliquetis	Yes / oui
Sometimes / parfois	I haven't noticed anything / je n'ai jamais preté	Yes / oui
Rarely / rarement	I haven't noticed anything / je n'ai jamais preté	Yes / oui
Sometimes / parfois	Knocking/ cognement, Grinding / frottement mé	Yes / oui
Sometimes / parfois	Knocking/ cognement, Grinding / frottement mé	Yes / oui
Sometimes / parfois	Knocking/ cognement, Grinding / frottement mé	Yes / oui
Sometimes / parfois	Knocking/ cognement, Squealing /grincement,	Yes / oui

BEFORE

AFTER

5. How often do you experience unusual engine sounds?/À quelle fréquence entende	6. Can you describe any sound your car has made before a breakdown?/Quel type de
Sometimes	Knocking
Sometimes	Knocking
Sometimes	Knocking
Sometimes	Knocking
Sometimes	Grinding
Sometimes	Grinding
Sometimes	Grinding
Sometimes	Grinding
Sometimes	Clicking
Sometimes	Clicking
Sometimes	Clicking
Sometimes	Clicking
Rarely	Squealing
Rarely	Squealing
Rarely	Squealing
Rarely	Squealing
Rarely	Squealing
Rarely	Squealing
Rarely	Squealing

- **Handling of Multiple-Selection Responses**

  Several questions in both surveys allowed respondents to select more than one answer. Question 9 of Survey 1, which asked respondents to identify desired application features, and Questions 1, 4, 5, 7, 9, 11, and 12 of Survey 2 were all multi-select questions. In the raw Google Forms export, such responses are stored as a single comma-separated string per respondent, rather than as individual values. During cleaning, each multi-select response string was parsed and disaggregated so that every selected option was counted individually. This allowed accurate feature-level frequency analysis — for example, correctly determining that 25 out of 30 car owner respondents selected "Explain warning lights" as a desired feature, even though many of those respondents selected it as part of a longer combined response string.

  BEFORE(multiple choice answers separated by commas)

L
9. What features would you like in such an app?/Quelles fonctionnalités aimeriez-vous dans une telle application ?
Explain warning lights /Explication des voyants du tableau de bord, Identify engine sounds/ identification de bruits de moteur, Cos
Explain warning lights /Explication des voyants du tableau de bord
Explain warning lights /Explication des voyants du tableau de bord, Identify engine sounds/ identification de bruits de moteur, Ste
Explain warning lights /Explication des voyants du tableau de bord, Connect to a mechanic/ contacter un mechanicien, Access of
Explain warning lights /Explication des voyants du tableau de bord, Identify engine sounds/ identification de bruits de moteur, Cos
Explain warning lights /Explication des voyants du tableau de bord, Identify engine sounds/ identification de bruits de moteur, Vid
Explain warning lights /Explication des voyants du tableau de bord, Step-by-step repair guides / Guides de réparation étape par
Explain warning lights /Explication des voyants du tableau de bord, Step-by-step repair guides / Guides de réparation étape par
Step-by-step repair guides / Guides de réparation étape par étape, Connect to a mechanic/ contacter un mechanicien
Explain warning lights /Explication des voyants du tableau de bord, Identify engine sounds/ identification de bruits de moteur
Step-by-step repair guides / Guides de réparation étape par étape

9. What features would you like in such an app?/Quelles fonctionnalités aimeriez
Identify engine sounds
Step-by-step repair guides
Video Tutorial
Connect to a mechanic
Cost estimates for repairs
Access offline
Explain warning lights
Identify engine sounds
Step-by-step repair guides
Video Tutorial
Connect to a mechanic
Cost estimates for repairs
Access offline
Explain warning lights
Explain warning lights
Connect to a mechanic
Connect to a mechanic
Access offline
Access offline
Explain warning lights

- **Removal of Irrelevant and Ambiguous Open-Text Entries**
  Both surveys included open-text fields to capture responses under "Other" options and optional explanatory questions. During cleaning, these free-text entries were reviewed individually. Entries that were clearly irrelevant, unintelligible, or constituted non-responses — such as single-character entries, placeholder text, and responses such as "No", "None", "Kilo hadison", and "." — were excluded from the qualitative analysis. Meaningful open-text responses, such as those explaining why a respondent would or would not trust the application, were retained and categorised thematically to supplement the quantitative findings.
- **Normalisation of Response Formatting**
  Minor formatting inconsistencies were identified and corrected across both datasets. These included leading and trailing whitespace in response strings, inconsistent capitalisation, and slight textual variations in otherwise identical answers introduced by different browsers or input methods. These were standardised to ensure that automated frequency counting produced accurate results without artificially splitting identical answers into separate categories.

```
= Table.Distinct(#"Changed Type")
```

  The above formular was used for duplicate rows
- **Verification of Completeness**
  All required questions in both surveys were marked as mandatory in Google Forms, which prevented respondents from submitting incomplete forms for those fields. As a result, no missing values were found in the required questions of either dataset. Optional fields — specifically the "Why?" explanation field in Question 8 of Survey 1 — contained blank entries for respondents who chose not to answer, and these were

correctly treated as non-responses and excluded from the qualitative analysis of that field without affecting the overall response count.
To achieve this we used the formular in a power query section of data ribbon in excel

```
= Table.ReplaceValue(#"Changed Type","",null,Replacer.ReplaceValue,{"1.Do you understand all the
    warning lights on your car dashboard? #(lf)Comprenez-vous tous les voyants du tableau de bord de
    votre voiture ?#(lf)"})
```

BEFORE

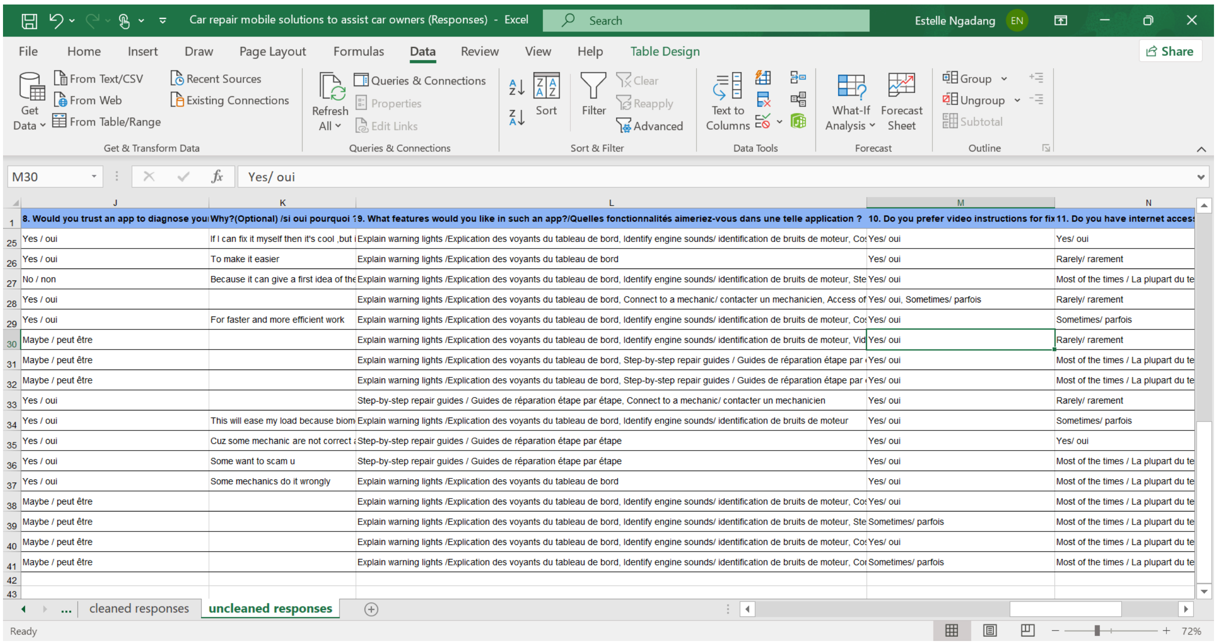
J	K	L	M
n app to diagnose you	Why?(Optional) /si oui pourquoi ?	9. What features would you like in such an	10. Do you prefer video i
	If I can fix it myself then it's cool ,but i	Explain warning lights /Explication des voyants	Yes/ oui
	To make it easier	Explain warning lights /Explication des voyants	Yes/ oui
	Because it can give a first idea of the	Explain warning lights /Explication des voyants	Yes/ oui
		Explain warning lights /Explication des voyants	Yes/ oui, Sometimes/ parfoi
	For faster and more efficient work	Explain warning lights /Explication des voyants	Yes/ oui
		Explain warning lights /Explication des voyants	Yes/ oui
		Explain warning lights /Explication des voyants	Yes/ oui
		Explain warning lights /Explication des voyants	Yes/ oui
		Step-by-step repair guides / Guides de réparat	Yes/ oui
	This will ease my load because biom	Explain warning lights /Explication des voyants	Yes/ oui
	Cuz some mechanic are not correct	Step-by-step repair guides / Guides de réparat	Yes/ oui
	Some want to scam u	Step-by-step repair guides / Guides de réparat	Yes/ oui
	Some mechanics do it wrongly	Explain warning lights /Explication des voyants	Yes/ oui
		Explain warning lights /Explication des voyants	Yes/ oui
		Explain warning lights /Explication des voyants	Sometimes/ parfois

AFTER

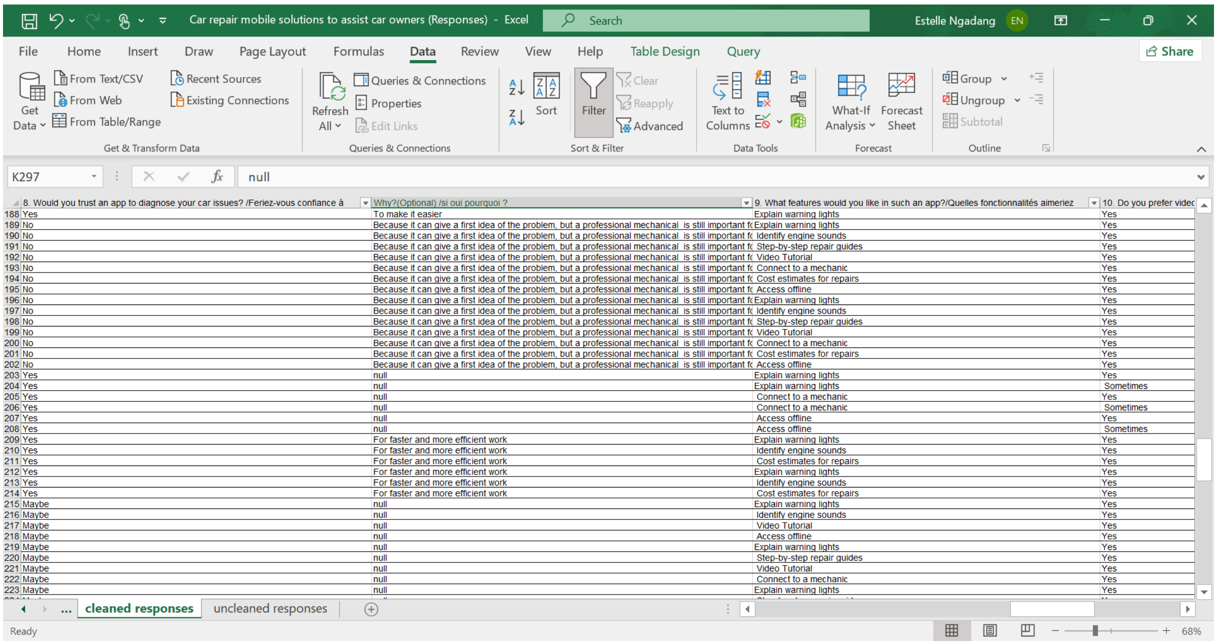
Why?(Optional) /si oui pourquoi ?	9
Because it can give a first idea of the problem, but a professional mechanical is still important for confi	I
Because it can give a first idea of the problem, but a professional mechanical is still important for confi	S
Because it can give a first idea of the problem, but a professional mechanical is still important for confi	V
Because it can give a first idea of the problem, but a professional mechanical is still important for confi	C
Because it can give a first idea of the problem, but a professional mechanical is still important for confi	C
Because it can give a first idea of the problem, but a professional mechanical is still important for confi	A
Because it can give a first idea of the problem, but a professional mechanical is still important for confi	E
Because it can give a first idea of the problem, but a professional mechanical is still important for confi	I
Because it can give a first idea of the problem, but a professional mechanical is still important for confi	S
Because it can give a first idea of the problem, but a professional mechanical is still important for confi	V
Because it can give a first idea of the problem, but a professional mechanical is still important for confi	C
Because it can give a first idea of the problem, but a professional mechanical is still important for confi	C
Because it can give a first idea of the problem, but a professional mechanical is still important for confi	A
null	E
null	E
null	C
null	C
null	A
null	A
For faster and more efficient work	E
For faster and more efficient work	I
For faster and more efficient work	C
For faster and more efficient work	E
For faster and more efficient work	I
For faster and more efficient work	C
null	E
null	I
null	V
null	A

- **Outcome of the Data Cleaning Process**
  The cleaned dataset formed the reliable foundation upon which the data analysis in the following section is based. All percentage figures, frequency counts, and feature rankings cited throughout this report are derived from the cleaned and consolidated data. Furthermore, the cleaning process itself revealed several data quality considerations that directly informed non-functional requirements for the system — most notably the need for clear, simple, and unambiguous language within the application interface, given that bilingual and formatting inconsistencies in the survey data reflected the broader linguistic diversity of the target user population in Cameroon.

BEFORE



AFTER



4. USER RELUCTANCE

While the survey data indicates a strong interest in a mobile diagnostic solution, it also revealed a significant "Trust Gap." Understanding why users are hesitant is crucial for designing a system that people will actually rely on during a vehicle crisis.

4.1 The Accuracy-Trust Paradox

The primary source of reluctance stems from the fear of **misdiagnosis**.

- **User Concern:** Car owners are worried that the app might misinterpret a sound or a light, leading to "false confidence" or unnecessary panic.
- **The "Human" Factor:** Many users still prefer the physical presence of a mechanic who can "feel" the car's vibrations, a sensation they feel a mobile app cannot replicate.

4.2 Technical Intimidation and Hardware Doubts

A significant portion of the target demographic expressed concerns regarding their own devices:

- **Microphone/Camera Reliability:** Users doubt if a standard smartphone microphone can distinguish between a "knock" and background street noise in a busy Cameroonian city.
- **Digital Literacy:** Some users are reluctant to adopt the tool because they fear the interface might be too technical or "too smart" for them to navigate during a stressful breakdown.

4.3 Safety and Liability Concerns

There is a notable fear regarding the "Tutorial" aspect of the project:

- **Fear of "Breaking it Further":** Users are reluctant to follow step-by-step guides for minor repairs because they lack the confidence to touch engine components, fearing they might cause more expensive damage.
- **Lack of Accountability:** Unlike a physical garage where a mechanic is accountable for a repair, users feel "on their own" when using an app, leading to hesitation in following its advice.

4.4 Environmental and Infrastructural Barriers

The Cameroonian context introduces specific reluctance related to infrastructure:

- **Data Costs and Connectivity:** Users are reluctant to rely on a tool that might require a heavy internet connection to process audio or video, especially when they are in remote areas with poor signal.
- **Bilingual Confusion:** If the terminology used in the app doesn't align with local "mechanic slang" (in either English or French), users may find the results untrustworthy or confusing.

5. CONCLUSION

The requirement gathering and data cleaning phases for **Task 2** have provided a foundational roadmap for the development of a diagnostic mobile application tailored to the Cameroonian automotive context. By bridging the gap between the technical expertise of mechanics and the practical challenges faced by car owners, the following key conclusions have been established:

5.1 Synthesis of Findings

The study confirms a significant **knowledge gap** among car owners, with over 60% unable to interpret critical dashboard warnings. This vulnerability is compounded by a reliance on informal advice or a tendency to ignore faults until they become catastrophic. Conversely, the expert survey validated that many common engine failures are preceded by detectable auditory and visual cues, such as knocking or specific warning lights. This alignment confirms that a mobile solution—utilizing **Audio**

Classification and **Image Recognition**—is not only feasible but necessary to improve vehicle longevity and road safety.

5.2 Strategic System Decisions

Based on the evidence gathered, the proposed system will pivot toward a **"Hybrid Assistance"** model:

- **Audio-Visual Diagnostics:** The app will simulate a mechanic's "eyes and ears" by using the smartphone's microphone for sound mapping (knocking, squealing) and the camera for warning light identification.
- **Safety-First Tutorials:** To mitigate the risk identified by mechanics regarding "complex DIY fixes," the app will restrict its guidance to basic maintenance (oil, batteries, tires) while flagging professional intervention for critical engine or brake failures.
- **Offline Accessibility:** Given the technical constraints identified in the user survey, integrating **offline AI functionality** is a non-functional requirement that will ensure the app remains functional in areas with poor internet connectivity.

5.3 Data Integrity and Reliability

The rigorous **Data Cleaning** process—addressing bilingual inconsistencies and multi-selection complexities—ensures that the system's logic is built on a statistically sound dataset. The transformation of raw, ambiguous entries into normalized data points has allowed the team to prioritize features that directly address user reluctance, specifically the need for high-accuracy results and simple, jargon-free instructions.

5.4 Final Outlook

In conclusion, the research demonstrates that while users are open to mobile car diagnostics, their "moderate trust" necessitates a system that prioritizes **reliability and transparency**. By incorporating mechanic-validated sound-fault mappings and focusing on the most misunderstood warning lights, the resulting application will serve as a vital intermediary tool. It will empower users with immediate knowledge while respecting the boundaries of professional mechanical repair, ultimately fostering a more informed and proactive car maintenance culture.